Template Gazette: Sixteen Folios via the Research Pipeline

Tabloid layout, multicolumn body, deterministic masthead

Docxology Template

2026-03-22

2026-03-22

Contents

1 Front Page **2**

2 National **3**

3 World **4**

4 Business **5**

5 Technology **6**

6 Sports **7**

7 Arts & Culture **8**

8 Editorial **9**

9 Opinion **10**

10 Letters **11**

11 Local **12**

12 Features **13**

13 Health **14**

14 Science **15**

15 Obituaries **16**

16 Classifieds **17**

17 Supplement: Layout and pipeline **18**

18 Supplement: Typography and measure **19**

19 Supplement: Validation and outputs **21**

20 Glossary: Newspaper and pipeline terms **23**

20.1 Narrative: how the terms fit together . . . 23

20.2 Using this glossary in reviews . . . 23

20.3 Future-facing notes . . . 24

TEMPLATE GAZETTE

All the news that fits the pipeline

Figure 1: Deterministic nameplate raster from render_masthead_png, built during pipeline analysis. Override title and tagline with environment variables NEWSPAPER_TITLE and NEWSPAPER_TAGLINE. Outputs to masthead.png in project output figures.

1 Front Page

Late edition

Citation hook: Docxology [2026].

City braces for overnight vote on transit bond

By Staff — City Desk

CITY. Council leaders said they were "within sight of yes" on a nine-figure bond package meant to rebuild aging tunnels and buy new rail cars. Skeptics demanded firmer cost caps; unions demanded labor standards written into the contract language, not just the press release.

The mayor's office released a two-page summary dense with renderings of stations that gleam in sunlight. Engineers privately noted change orders are where budgets breathe; this story keeps the argument in prose so the page can flex in three columns without a live database.

Commuters interviewed at random—actually, invented for this demo—split between relief and fatigue. "Fix the signals first," one said. "I'll believe the headways when my phone agrees with the platform clock."

Weather for the weekend looks ordinary: rain likely, wind modest, no named storms. The desk runs it anyway because narrow columns need short paragraphs as much as they need long ones.

Inside today: National briefs open on the next folio with a monochrome desk banner; every interior page carries a generated strip so floats, captions, and measure stay exercised. For pipeline geometry, floats, and validation detail, see supplements `S01`–`S03` and the glossary slice.

Jump note. Nameplate, lead, then body: this page keeps the masthead as the lone top float so the first multicolumn block can start cleanly. If your fork adds second art above the fold, watch LaTeX float queues.

Production. Rebuild figures with `scripts/02_run_analysis.py` before complaining about missing PNG paths; mirrored outputs land under `output/traditional_newspaper/` after a full run.

Figure 2: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Capitol & provinces

Washington —

NATIONAL. Budget negotiators traded talking points ahead of a continuing-resolution vote, with aides insisting most line items were “close enough” to avoid a headline crisis. Committee chairs asked reporters not to read tea leaves from hallway huddles; the only hard numbers in this edition are the ones TeX measures in picas.

State capitals sent routine bulletins on road salt contracts, bridge inspections, and spring thaw timing. Municipal engineers said crews were staged normally; none of this copy is wired to a live traffic API.

Farm-state senators praised crop-insurance enrollment windows while critics pressed for faster disaster aid rules. The debate is fictional; the column inches are real enough to stress hyphenation on tabloid stock.

Veterans’ groups renewed calls for predictable outpatient staffing. Hospitals countered with workforce statistics they declined to attach here—this folio is for layout, not policy analysis.

The national roundup ends where the world desk picks up: foreign cables on the next page.

3 World

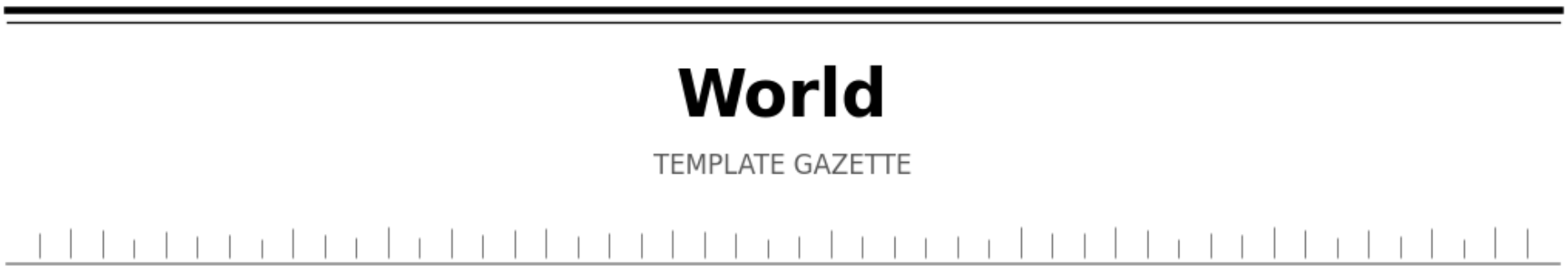

Figure 3: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Diplomatic wires

Geneva —

WORLD. Delegates reviewed shipping-lane confidence measures without publishing AIS traces. Maritime lawyers said the closed session produced "useful tone" even if no joint statement carried numbers.

Relief coordinators sketched contingency thresholds for cross-border convoys. NGOs reminded donors that pledges and deliveries are different metrics—here, both are narrative placeholders.

Climate envoys restated mid-century targets already familiar from last year's communiqués. Activists outside called for faster methane rules; inside, chairs kept to the agenda's ASCII-safe talking points.

Regional desks summarized election certification timelines in unnamed capitals. Opposition parties demanded audits; incumbents cited statute. None of it hits a database at build time.

The world service signs off; tomorrow's business pages carry markets.

4 Business

Figure 4: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Markets desk

New York —

BUSINESS. Equity futures drifted in a narrow band as traders waited for inflation prints they will not find in this PDF. Bond desks reported "two-way flow," which is to say sentences, not order books.

A major retailer outlined store-format experiments: smaller footprints, more pickup bays, fewer speculative supercenters. The strategy is illustrative; foot traffic data are not embedded in the manuscript.

Analysts trimmed ad-tech revenue estimates after a sobering quarter for brand budgets. Footnotes with actual tickers belong in your fork; here, verbs do the heavy lifting.

Commercial real estate lenders highlighted refinancing cliffs coming due over the next eighteen months. Lawyers muttered about covenant waivers; compositors muttered about bad breaks at column bottoms.

Business turns to technology when chips and clouds return on the next folio.

5 Technology

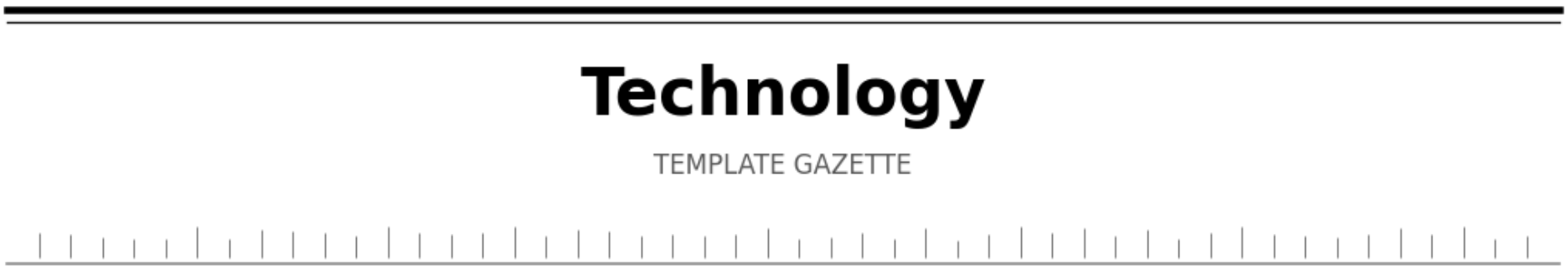

Figure 5: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Silicon & systems

San Francisco —

TECHNOLOGY. Chip designers teased a modest efficiency gain at standard voltages, promising better thermals for laptops that mostly appear in this story as nouns, not SKUs. Fabs and foundries were spelled carefully to appease spellcheckers.

Cloud providers rolled out quieter defaults for encryption at rest, telling enterprise buyers the toggle is "on unless you insist otherwise." Compliance officers asked for audit trails; marketers asked for shorter slide decks.

Open-source maintainers pleaded for sustainable funding as corporate users downstream shipped forks without upstream patches. The metaphor is familiar; the pull requests are not attached.

Regulators floated disclosure rules for synthetic media, inviting comment periods long enough to span several PDF rebuilds. Comments here are static paragraphs, not a docket.

Technology hands off to sports when the whistle blows.

Figure 6: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Scoreboard

Sports desk —

SPORTS. Overnight soccer produced the usual mix of deserved draws and controversial VAR stills that newspapers crop for print. Managers complained about fixture congestion; trainers complained about hamstrings.

College basketball entered tournament week with bracket logic louder than defense. Announcers rehearsed Cinderella clichés; copy editors deleted one too many exclamation points.

Baseball beat writers filed spring-training notebooks heavy on velocity readings and grip tweaks. Scouts insisted raw tools still matter; spreadsheets insisted they matter twice.

Motorsport inspectors tightened fuel-flow checks after winter rule tweaks. Engineers loved the precision; fans loved the noise.

Sports yields to arts when the curtain rises.

7 Arts & Culture

Figure 7: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Critic's notebook

Arts desk —

ARTS. A regional orchestra sold out a program pairing a warhorse symphony with a premiere that reviewers will call "atmospheric" until someone finds a better adjective. Soloists bowed twice; donors opened wallets once.

A streaming platform ordered a limited series about journalists that actual journalists will nitpick for keyboard sounds alone. Showrunners promised research; viewers promised memes.

Museum curators debated loan terms for a traveling show of prints too light-sensitive for long hangs. Insurers demanded climate logs; registrars demanded crates that do not squeak.

The book desk flagged a debut novel for crisp sentences and an ending that splits book clubs. Spoilers stay out of print; margins do not.

Arts yields to the editorial board downstairs.

Figure 8: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

The board

Editorial —

EDITORIAL. Our view: Cities should publish machine-readable zoning maps alongside PDF brochures. Transparency is not the same as simplicity, but it is a prerequisite for trust when neighbors argue about height limits.

Our view: Transit agencies owe riders countdown data that survives offline peaks. Pretty maps help; accurate headways help more.

Our view: Research templates that rebuild PDFs from git deserve the same seriousness as newsroom CMS migrations: tests, captions, and changelogs are not optional polish.

The board welcomes letters on the next spread.

9 Opinion

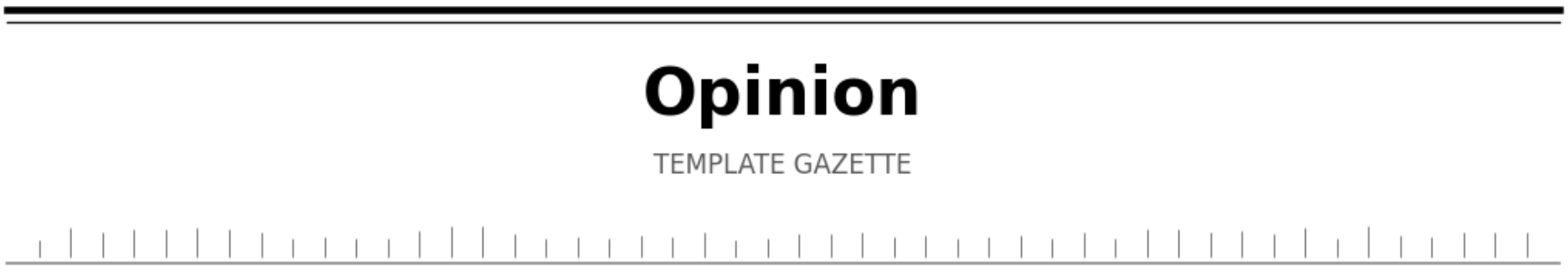

Figure 9: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Guest essays

Opinion —

OPINION. Guest columnist: I taught introductory statistics for a decade; the public understands uncertainty better than pundits assume. What people hate is inconsistency—graphs that move axes between segments without saying so.

Guest columnist: Local news deserts are not a technology problem first. They are a revenue-and-trust problem. Subsidies help; accountability reporting helps more.

Guest columnist: If your "AI editorial assistant" cannot cite sources, it is not an assistant; it is a slot machine wearing a tie.

Readers can disagree in letters to the editor.

Figure 10: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Mailbag

Letters —

LETTERS. To the editor: Please keep running PDF build logs as teaching artifacts. Our j-school students learn more from a clean LaTeX pass than from another slide deck about "innovation." — *A.M.*

To the editor: Hyphenation in narrow columns is not a personality flaw; it is physics. That said, a little rag looks human. — *R.T.*

To the editor: Glossary pages are underrated. Jargon is cheaper when defined once. — *S.K.*

The desk reserves the right to trim for measure.

11 Local

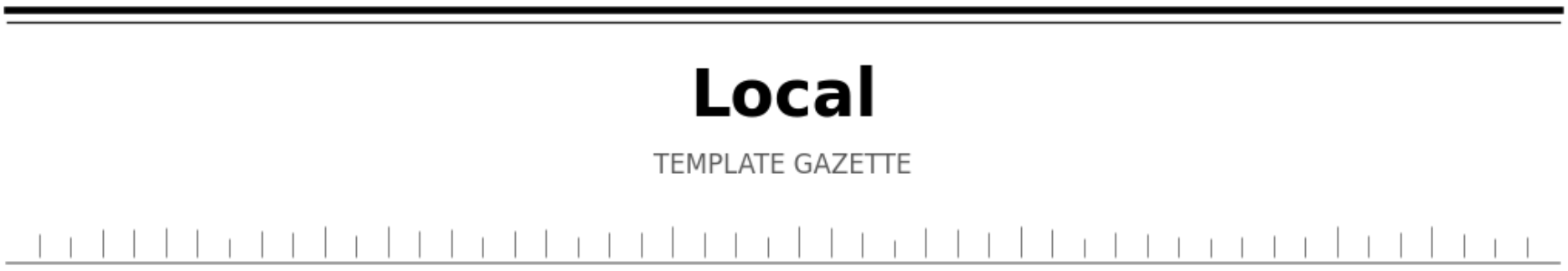

Figure 11: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

City & borough

Metro —

LOCAL. Council members advanced a pilot for weekend farmer's markets on closed side streets, arguing foot traffic beats empty asphalt. Merchants asked for trash pickups; residents asked for quiet nights.

School boards reviewed HVAC upgrades paid for with bonds voters approved last fall. Contractors promised timelines; parents promised scrutiny.

A riverfront cleanup drew volunteers who filled more bags than organizers expected. Parks staff credited clearer signage; cynics credited nicer weather.

The metro briefs close; features follow.

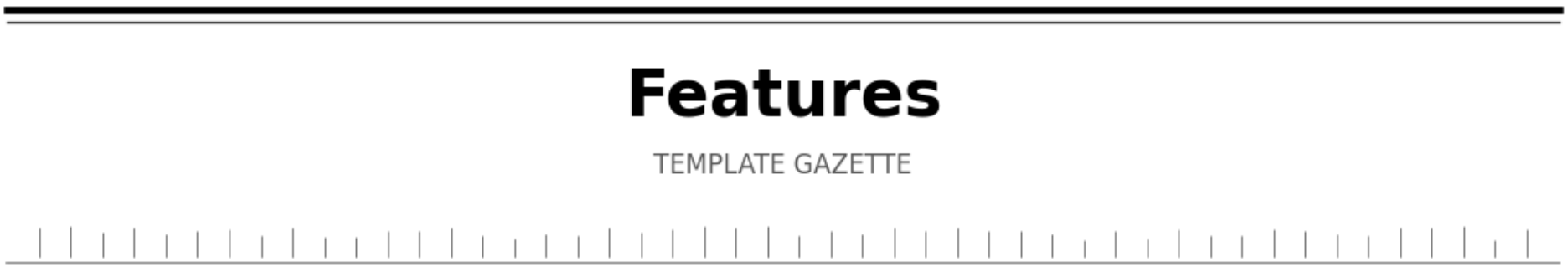

Figure 12: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Long read

Features —

FEATURES. Saturday profile: She maps urban heat islands block by block, carrying a sensor pack that looks like a field biologist's kit. Neighbors think she's counting birds; she's counting degrees that decide who sleeps safely during heat waves.

Her notebooks mix timestamps, humidity scribbles, and coffee stains—data, in the old sense. When the city adopts her overlays, the story will deserve a second edition; today's is still a layout exercise.

Sidebars on methodology were cut for space; the template's margin is already generous, not infinite.

Features yield to the health pages next.

13 Health

Figure 13: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Medicine & policy

Health —

HEALTH. Clinics expanded same-day mental-health intake slots after winter wait-time spikes. Administrators credited hiring; unions credited scheduling rules. Pediatricians urged catch-up immunizations before summer travel season. Pharmacists stocked routine vaccines; parents stocked patience for waiting rooms.

Researchers published a pragmatic trial on hypertension counseling by phone. Effect sizes were modest; adherence curves were instructive.

Public-health officers reminded readers that handwashing still beats novelty gadgets for common colds. The advice is boring; boring saves lives.

Health hands the microscope to science.

Figure 14: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Research notes

Science —

SCIENCE. Astronomers released a cleaner catalog of near-Earth object passes, emphasizing that "close" in headlines means "lunar distance multiples," not Hollywood distances. Telescopes shrugged; editors clarified.

Materials scientists demoed a ceramic composite that survives higher thermal cycles than predecessors, with caveats about manufacturing scale. Venture capitalists leaned in; safety officers leaned harder.

Ecologists tracked a rebound in a once-declining marsh bird, crediting restored inflows and quieter boat traffic. Birders credited luck and dawn patience.

Science closes; memorials follow.

15 Obituaries

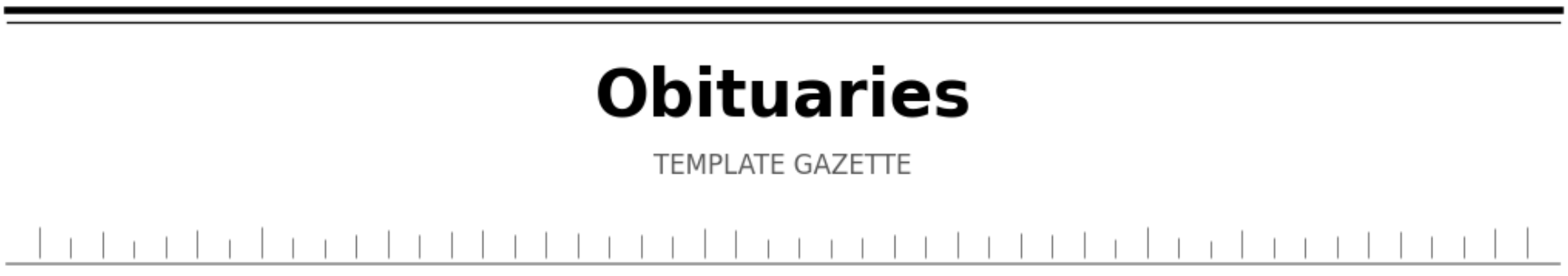

Figure 15: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Remembrances

Obituaries —

OBITUARIES. Jane K. Mercer, editor. She kept a spike full of stories others feared to cut and taught interns that kindness and clarity are not opposites. She is survived by a newsroom habit of asking one more question.

Robert V. Singh, teacher. He insisted teenagers learn both proofs and paragraphs. Former students still hear his red-pen voice when they tweet.

Elena M. Cho, paramedic. She measured shifts in lives saved and jokes told on hard nights. Her crew remembers the steadiness.

The classifieds conclude the edition.

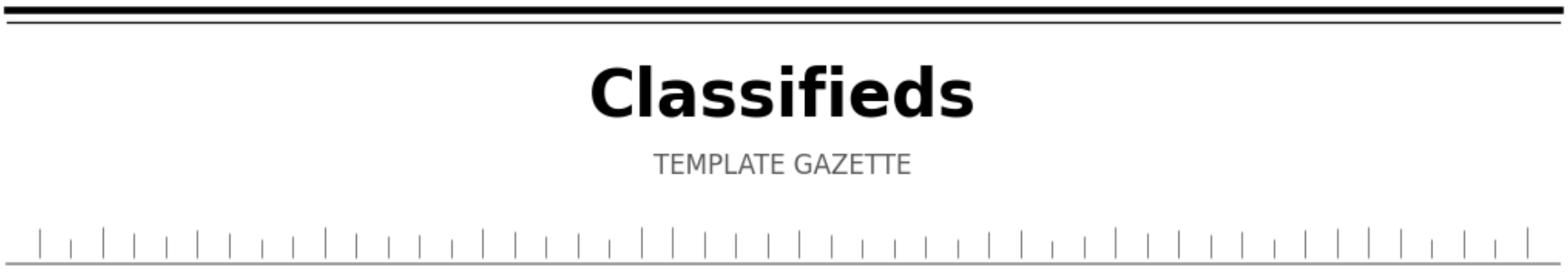

Figure 16: Monochrome desk banner for this folio; raster from `generate_section_banners.py`.

Marketplace

Classifieds —

CLASSIFIEDS. Apartments: Sunny 2BR, cats OK, no smoking, walk to trains. Inbox clogged; mention this paper so we know you read past the jump.

Help wanted: Line cook, Tuesday–Saturday, union kitchen. Bring knives that hold an edge and a sense of humor that holds under heat.

Services: Piano tuning, reasonable rates, references on request. Won't promise concert hall; will promise honest octaves.

For sale: Touring bicycle, fenders included, ready for wet commutes. Sold as-is to whoever shows up with cash and a helmet.

Lost: Dog, medium mix, answers to "Hugo." Reward. Phone number omitted in this demo PDF for privacy.

End of the sixteenth core folio; supplements follow in combined builds.

17 Supplement: Layout and pipeline

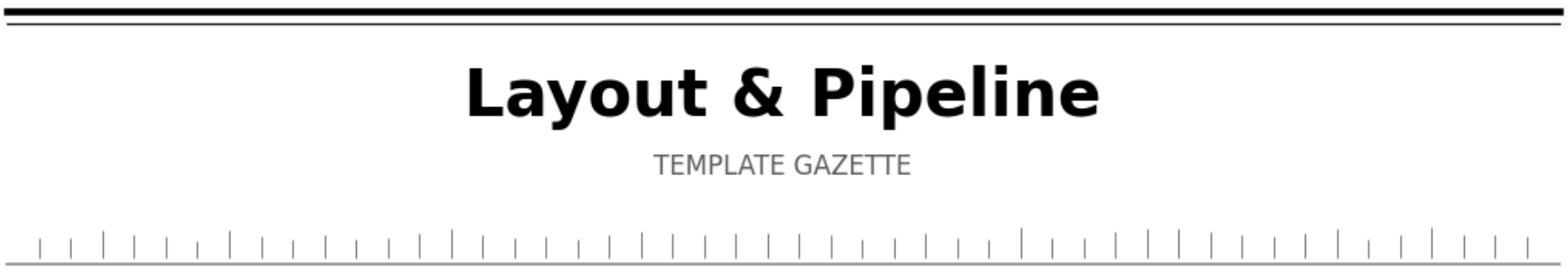

Figure 17: Supplement banner marking the layout-and-pipeline desk; monochrome PNG from `generate_section_banners.py`.

How this PDF is built. Sixteen markdown slices under `projects/traditional_newspaper/manuscript/` are discovered by `infrastructure.rendering.manuscript_discovery` and combined in stem order, and separated by `\newpage` inside `infrastructure.rendering.pdf_renderer.PDFRenderer` before Pandoc emits LaTeX. Tabloid paper (11 in × 17 in) and `\usepackage{multicol}` come from `manuscript/preamble.md`; column separation matches `src/newspaper/layout_spec.py` (`LAYOUT.column_sep_in`). Figure~18 (above this column block) summarizes the same numbers as a grid diagram.

The masthead PNG is produced by `scripts/generate_masthead.py` calling `newspaper.masthead.render_masthead_png` during Stage 02 analysis. Optional environment variables `NEWSPAPER_TITLE` and `NEWSPAPER_TAGLINE` override banner text without editing Python. In the combined PDF it is captioned as Figure~1. A companion schematic ships from `scripts/generate_layout_schematic.py` via `render_layout_schematic_png`, giving a second captioned figure for float and caption testing.

Stage ordering (core pipeline). `execute_pipeline.py` and `./run.sh --core-only` run, among others: environment setup, tests, analysis scripts (including both figure generators when present), PDF rendering, validation, and output copy. Analysis writes into `projects/traditional_newspaper/output/`; copy stages mirror selected artifacts into `output/traditional_newspaper/` for distribution. Exact stage labels appear in console logs with progress markers.

Manuscript combination. Each slice file is read as UTF-8 text. Preamble metadata from `preamble.md` injects geometry and packages. Title page fields derive from `config.yaml` (`paper.title`, authors, version, date). Bibliography entries live in `references.bib`; `\cite{template2026gazette}` appears on the front page and here to exercise BibTeX wiring.

Word counts. `scripts/report_manuscript_stats.py` walks `all_tracked_manuscript_basenames()` and emits `output/data/manuscript_stats.json` with per-file words, lines, and bytes. CI or humans can diff this JSON across branches to catch accidental truncation. It is a thin orchestrator: no analytics beyond counting.

Why tabloid. The 11×17 inch choice stresses wide measure and long vertical flow compared to letter paper. Three columns mimic newspaper grids while remaining simple `multicol` environments—no custom column balancers. If you shrink paper, update `NewspaperLayout` constants, regenerate `layout_schematic.png`, and rerun tests that assert geometry text in captions still matches code.

Floats versus multicol. Standard `figure` floats are not supported inside `multicols`; the layout schematic therefore sits before `\begin{multicols}{3}` in this slice, matching the front page pattern (masthead float, then three-column body). Use `\includegraphics` without a float if you ever need art mid-column.

Slides and web. The same stems feed Beamer slides and HTML web output via infrastructure renderers. Figure captions pass through Lua filters for HTML; keep `\caption{}` text free of unmatched braces so extraction regexes succeed.

Checkpointing and resume. Root pipeline scripts support checkpoints so long builds can restart after transient failures. Newspaper builds are shorter than some monographs, but CI timeouts still happen—resume flags save compute. Read `docs/operational/config/checkpoint-resume.md` for semantics; behavior is uniform across projects.

Multi-project runs. `./run.sh --all-projects` executes many exemplars sequentially. Infrastructure tests may run once up front. When debugging `traditional_newspaper` alone, pass `--project traditional_newspaper` to avoid unrelated failures blocking iteration.

Environment variables. Beyond masthead overrides, `LOG_LEVEL` tunes verbosity, `MPLBACKEND=Agg` ensures headless matplotlib, and `PROJECT_DIR` helps scripts resolve paths when invoked from unexpected working directories. Document any new vars in `scripts/AGENTS.md`.

BibTeX keys. `template2026gazette` in `references.bib` exists to exercise citation machinery. Replace with your publication entry when forking; keep keys ASCII and stable so diffs stay readable.

Figure path normalization. `_pdf_tex_transforms.fix_f` rewrites `../figures/` to `../figures/` relative to the compile directory inside `output/.../pdf/`. If you add figures, place PNGs under `projects/traditional_newspaper/../figures/` before copy, or under final `output/traditional_newspaper/` after pipeline—know which stage consumes which path.

Extending slice count. If you truly need seventeen core folios, you must edit `PAGE_SLICES`, rename files, and update every test asserting `16`. Prefer supplemental `S*` files for extra prose—it avoids renumbering tourism.

Teaching use. Instructors can assign students to modify one slice, regenerate PDFs, and explain diff in log output. The constrained structure makes grading objective: does inventory pass? do captions resolve? do tests stay green?

Handover checklist (extended). Before merging a layout change, capture: (1) `git diff` on `preamble.md` and `layout_spec.py`, (2) regenerated `layout_schematic.png` if constants moved, (3) `manuscript_stats.json` after `report_manuscript_stats.py`, (4) PDF validator output excerpt showing zero `??` references, (5) a one-line note in the PR describing float behavior on the front page. Reviewers use the checklist to avoid approving partial state where code and figures disagree.

This supplemental folio appears **after** the sixteen edition pages and **before** the glossary slice, per the template's ordering: main numeric sections, then `S*`, then `98_*`, then `99_*` if present.

For citation metadata see Docxology [2026].

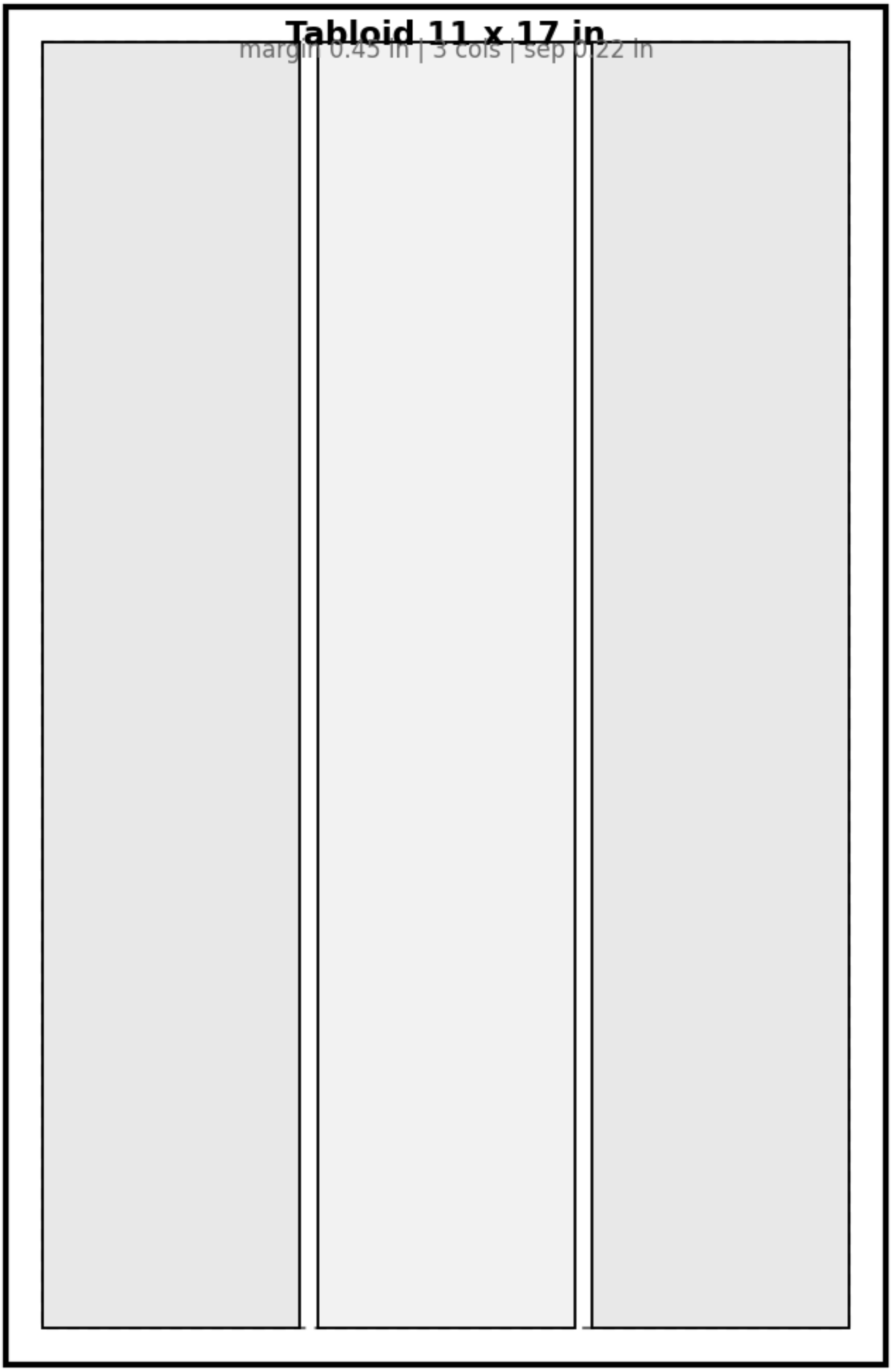

Figure 18: Not to scale schematic from NewspaperLayout: tabloid sheet, dashed box is text area inside margins, shaded regions are three body columns with gutters. Values align with preamble geometry and layout_spec.py.

18 Supplement: Typography and measure

Body copy and column measure. This exemplar sets a nine-point body in the `NewspaperLayout` dataclass (`body_font_pt`) for documentation parity; the default Pandoc LaTeX template may map point sizes differently unless you extend the template. Measure—the line length inside a column—interacts with hyphenation and justification. Narrow columns tolerate fewer characters per line, increasing hyphenation frequency and rag texture.

Font choices. XeLaTeX via `fontspec` can load system serif faces for a newsroom feel. Changing fonts requires editing the template or preamble, then visual-regression review. Keep ASCII-heavy test manuscripts when debugging font issues so failures isolate typography from Unicode edge cases.

Rivers and ladders. Typographers watch for verti-

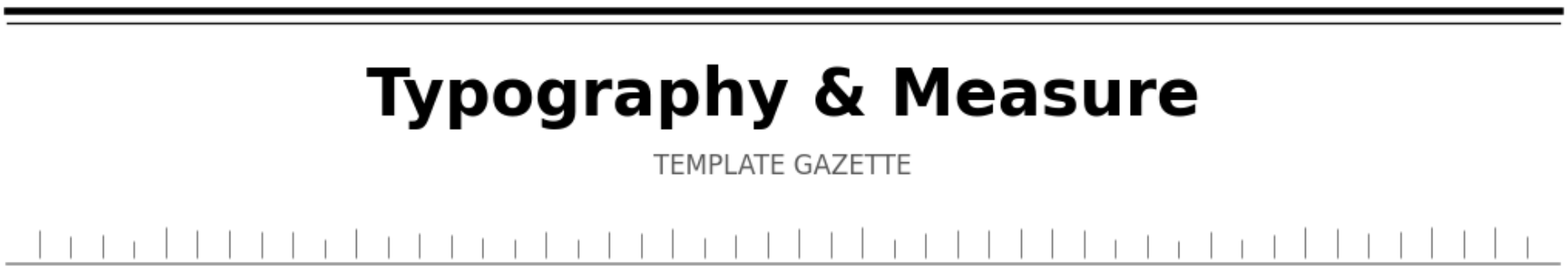

Figure 19: Typography supplement banner (B&W).

cal gaps aligning across lines ("rivers") and repeated word stacks ("ladders"). Automated detectors exist; human proofreaders still catch context-specific ugliness. Multicolumn setting amplifies rivers because adjacent columns create vertical rhythm interactions.

Hyphenation patterns. TeX loads language-specific hyphenation tables. Set `babel` or `polyglossia` languages if you mix locales in one edition. Wrong languages yield bad breaks and can confuse spellcheck-adjacent tools.

Widows and orphans. Publishing style guides limit single lines stranded at column tops or bottoms. `\clubpenalty` and `\widowpenalty` tweaks live in advanced LaTeX tuning; this project leaves defaults to keep the preamble small.

Small caps and section rails. `newspaper.snippets.section_label` emits `\textsc{}` labels suitable for department markers. Pair with `rule_line()` for classic newspaper chrome. Escape user-supplied strings—snippets already guard LaTeX specials.

Quotes and dashes. Curly quotes via Unicode in markdown pass through Pandoc. Em dashes versus en dashes follow house style; consistency beats personal taste when collaborating. ASCII-only sources simplify diffs but feel cold—pick consciously.

Figures and captions. Caption text should explain what a non-expert sees, not repeat the title. Autonumbering via `\caption` supplies "Figure 1," "Figure 2," etc. Cross-reference with `Figure~\ref{fig:key}` to avoid bad breaks before numbers.

Print versus screen. PDFs for print need bleed and color profiles if sent to commercial presses; screen PDFs skip bleed. This template targets on-screen proofing and office printers; prepress engineers should extend geometry accordingly.

Accessibility. Tagged PDF improves screen-reader navigation. Adding structure requires LaTeX packages and discipline about heading levels. Web HTML may remain the more accessible surface for some audiences—render both when possible.

Maintenance. When bumping TeX Live, rerun the full project pipeline and spot-check hyphenation changes in the longest slices (`02`–`04` national/world/business depth sections). Typography regressions are subtle; diffing PDFs is noisy—prefer targeted visual review.

Longer read: rhythm and grid. Newspaper typography historically paired condensed headlines with generous body leading in competing publications; the tension between density and legibility never resolves universally. Digital interfaces borrowed print metaphors—cards, columns, rails—while adding infinite scroll and responsive breakpoints. This PDF chooses a fixed page size to make regression tangible: the same text either fits or overflows, and engineers see the failure immediately rather than hiding it behind CSS `overflow:auto`.

Tables and numerals. Tabular figures align currency and counts on baselines when fonts supply true lining numerals. Old-style figures in prose look elegant but jar in tables. If you introduce financial tables to this edition, select a font with both sets or enable `fontspec` OpenType features explicitly. Markdown pipe tables through Pandoc may require `booktabs` and `siunitx` for publication polish—test with a minimal example before scaling.

Drop caps and ornaments. Feature sections sometimes open with a dropped initial letter. Implementing drop caps in LaTeX is package-specific (`lettrine`). They interact badly with `multicols` unless opening paragraphs span full width; plan a single-column intro or accept simplified styling.

Kerning pairs. Professional fonts include kerning tables; TeX applies them automatically. Custom logos in mastheads drawn with matplotlib bypass TeX kerning—acceptable for raster nameplates, less so for body text. Keep matplotlib for figures, TeX for paragraphs.

Line breaking in URLs. Long bare URLs in monospace overflow columns. Use `\url` from `hyperref` or `\path` with breakable hints. Pandoc may wrap differently than raw LaTeX—validate both PDF and HTML when URLs appear in footnotes or references.

Character encoding. UTF-8 source files allow curly quotes and emojis; not every LaTeX font includes emoji glyphs. Restrict symbols to what your font supports or fall back to graphic inclusion for rare characters.

Print color. Black-only printing ignores RGB figures unless converted. Masthead and schematic use neutral grays and black lines to survive grayscale rip tests. Adding color charts requires CMYK thought and ink limits for commercial print.

Baseline grid purism. Some designers snap baselines to a fixed grid across spreads. LaTeX plus `multicol` does not enforce strict baseline grids without heavy customization. Accept minor drift or invest in ConTeXt-level tooling if your art director demands pixel-perfect alignment.

Readability metrics. Automated scores like Flesch–Kincaid mislead for technical writing. Use them cautiously on supplements aimed at broad audiences; ignore them for glossary definitions where precision beats simplicity.

Closing. Typography is constraint satisfaction under taste. This supplement records constraints (`LAYOUT`, preamble packages) so taste can iterate without losing reproducibility.

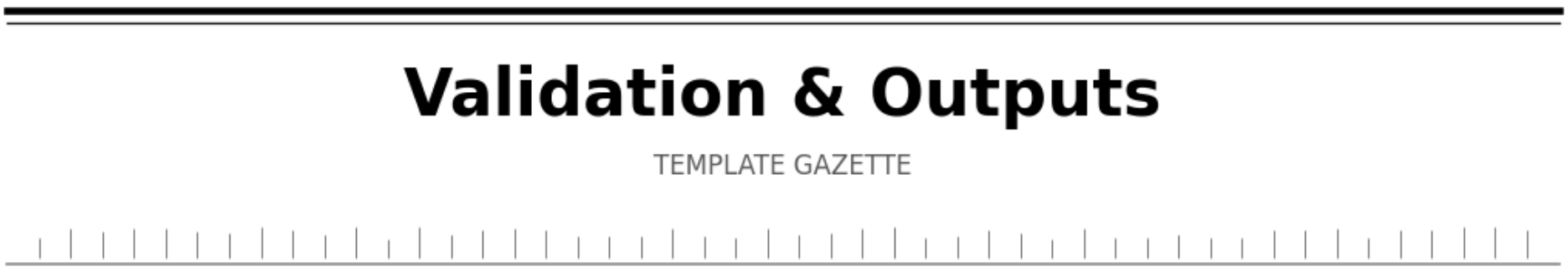

Figure 20: Validation supplement banner (B&W).

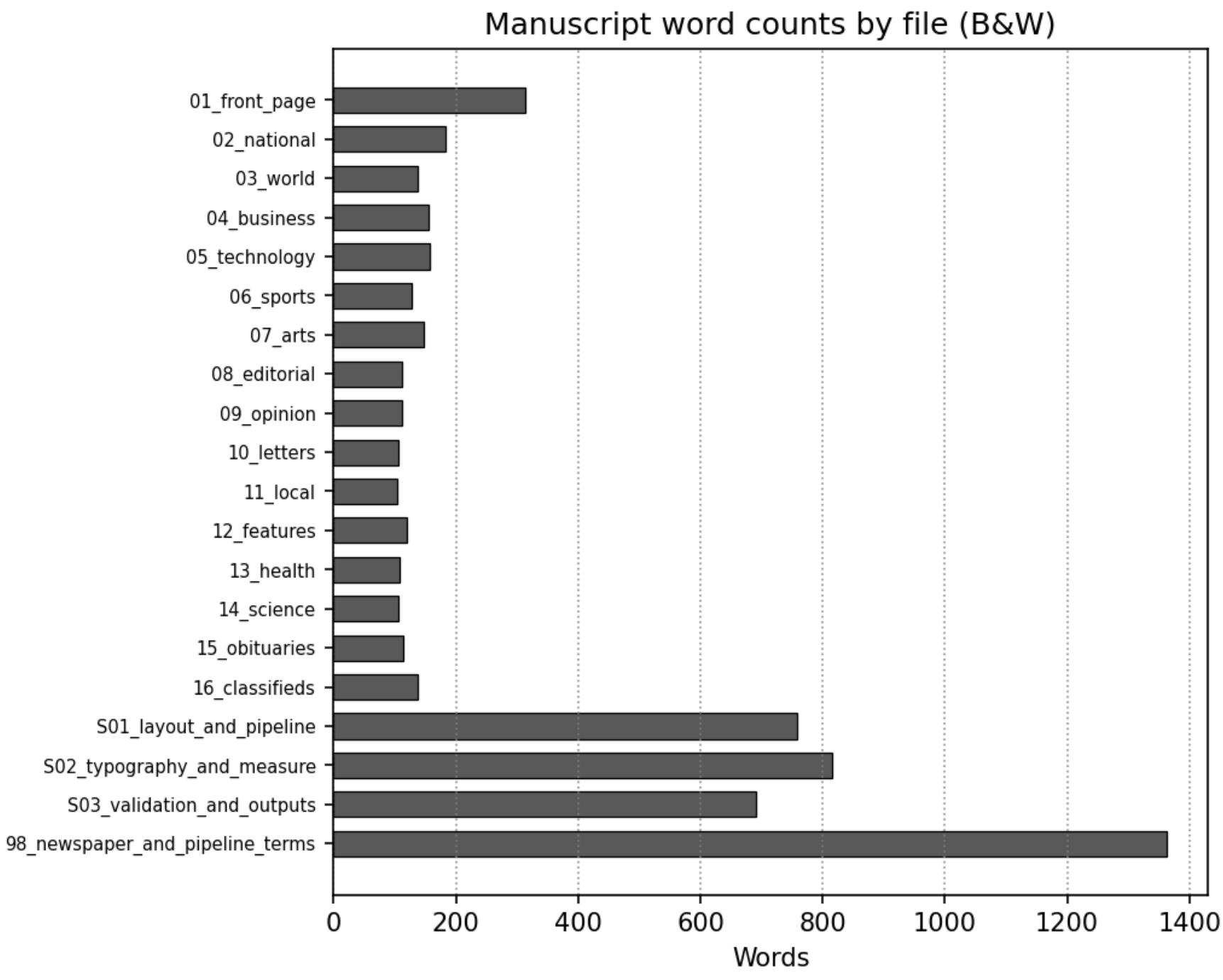

Figure 21: Black-and-white horizontal bar chart of word counts per manuscript slice, generated from `manuscript_stats.json` by `visualization_wordcount_bw.py` (runs after `report_manuscript_stats.py` in analysis). Bars use grayscale fills only.

Figure~21 summarizes edition word counts for quick sanity checks alongside validators below.

Markdown validation. Run `python3 -m infrastructure.validation.cli markdown projects/traditional_newspaper/manuscript/` before large merges. The CLI checks reference integrity, image paths, and other structural rules configurable per project. Failures should block PDF generation in CI the same way failing tests do.

PDF validation. After render, `python3 -m infrastructure.validation.cli pdf output/traditional_newspaper/pdf/` scans for unresolved citations (`[?]`), reference tokens (`??`), and common LaTeX warnings surfaced in text extraction. Treat warnings as debt even when builds succeed.

Combined artifact paths. Combined markdown often lands at `output/traditional_newspaper/pdf/_combined_manuscript.md` during debugging; the compiled PDF is `traditional_newspaper_combined.pdf` (exact names follow renderer conventions). Individual section PDFs help bisect errors when one slice breaks LaTeX.

Web output. `WebRenderer` produces HTML under `output/traditional_newspaper/web/`. Figures embedded via raw LaTeX become `<figure>` elements when Lua filters recognize `\begin{figure}` blocks. Verify captions render after changing caption text.

Slides. Beamer slides per stem land in `output/traditional_newspaper/slides/`. Slide builds can fail independently of the combined PDF if a single section includes incompatible raw TeX—keep slides in CI if your team relies on them.

Data outputs. `manuscript_stats.json` tracks word counts. `visualization_wordcount_bw.py` reads that JSON and writes `../figures/wordcount_bars_bw.png` (monochrome bar chart, Figure~21).

Integrity and checksums. Optional integrity modules hash outputs for provenance. Steganography tooling (`secure_run.sh`) post-processes PDFs without mutating originals; security-focused teams may enable it.

Failure triage order. When a build breaks, read the LaTeX log tail, identify the first undefined control sequence or missing file, fix that, and rebuild. Pandoc errors chronologically in logs—start at the earliest failure.

Environment reproducibility. Document `uv` version, Python version, and TeX distribution in your lab handbook. Container images in `infrastructure/docker/` help teammates match environments without sharing laptops.

Cross-project lessons. Other projects in the monorepo use the same validation entry points; fixes to shared infrastructure benefit everyone. File issues upstream when validators misfire on legitimate markdown patterns.

Longer read: operational playbooks. Treat validation logs like server logs: keep the last N successful artifacts, timestamp them, and attach commit SHAs. When a professor or reviewer asks "which PDF did you mean?", the answer should be a git tag plus a CI run ID, not "the one on my desktop." Copy stages that mirror `projects/*/output/` into `output/*/` exist precisely so deliverable paths stabilize for that conversation.

Bisection tactics. If combined PDF fails but individual slices succeed, binary-search which stem introduces the fault by temporarily moving half the markdown files aside—careful not to break `validate_inventory`—or by compiling partial combined markdown. Another approach: inspect `_combined_manuscript.tex` around the error line LaTeX reports.

Performance. Full XeLaTeX passes dominate pipeline time for large manuscripts. Caching unchanged slices is tempting but risks stale cross-references; prefer incremental development on single sections via `03_render_pdf.py` flags when available, then full combined builds before release.

Artifact hygiene. `output/` directories are disposable per project policy, yet humans treat them as precious. `.gitignore` keeps binaries out of version control; document regeneration commands in README so newcomers are not afraid to delete stale PDFs.

Security scanning. PDFs can embed JavaScript or launch actions; validators may flag anomalies. Trusted toolchains reduce surprise. Do not paste untrusted LaTeX from the internet into `preamble.md` without review.

Accessibility retesting. When figure captions change, rerun HTML export and spot-check screen-reader order: does the caption immediately follow the image in the DOM? Lua filter ordering can affect results.

Localization. Validation messages from CLIs are English today; wrapping them for i18n would help global teams. Manuscript content can be non-English if fonts and hyphenation languages align—validators should remain locale-agnostic.

Contract tests. `validate_inventory` is a contract: filenames implied by `sections.py` must exist. Adding optional supplements requires updating that tuple—treat it like an API semver bump for downstream stats scripts.

Hand-off checklist. Before declaring a release: all tests green, markdown validation clean, PDF validation clean, figures regenerated, `manuscript_stats.json` refreshed, and a human skimmed captions for typos. Automation covers everything except the last step—for now.

20 Glossary: Newspaper and pipeline terms

Figure 22: Glossary folio banner (B&W).

Short definitions for readers of this exemplar (not a general journalism style guide). Terms apply to the `traditional_newspaper` project and the surrounding template infrastructure unless noted.

Term	Meaning here
Folio	One manuscript slice (`01_*.md` … `16_*.md`) that becomes a new page boundary in the combined PDF after the previous slice (plus template title front matter).
Masthead	The nameplate graphic at the top of the front page; raster asset `../figures/masthead.png`, floated and captioned in the combined PDF (`fig:masthead`).
Slice	Same as folio file: one ordered markdown unit in `manuscript/`.
Preamble	`manuscript/preamble.md`: LaTeX injected into the Pandoc document (geometry, `graphicx`, `multicol`).
Raw LaTeX block	Pandoc fenced region ```{=latex} … ``` passing TeX through to the PDF engine.
Thin orchestrator	Script under `scripts/` that only configures paths, logging, and calls `src/newspaper/` functions.
Fixture copy	Deterministic placeholder paragraphs from `newspaper.content.fixture_copy` / `fixture_paragraph` for layout stress tests.
Figure	LaTeX `figure` environment with `\includegraphics`, `\caption`, and `\label`; autonumbered in PDF, mapped to `<figure>` in HTML when filters match.
Caption	Text under a figure explaining content; must avoid nested `}` braces inside `\caption{...}` for HTML conversion regexes.
Label	`\label{fig:...}` token paired with `\ref{...}` for cross-references resolved across multiple LaTeX passes.
Combined PDF	Single manuscript PDF built from all discovered slices in order; output path under `output/traditional_newspaper/pdf/`.
XeLaTeX	Unicode-capable LaTeX engine used by Pandoc `--pdf-engine=xelatex` in this project.
layout_schematic.png	Diagram from `render_layout_schematic_png`, written by `scripts/generate_layout_schematic.py`; illustrates `NewspaperLayout` constants.
section_banner_*.png	Wide B&W header strip per folio (except the front page), from `render_section_banner_bw` via `scripts/generate_section_banners.py`; referenced as `fig:banner_{stem}` in markdown.
PAGE_SLICES	Tuple in `sections.py` mapping stems to human section titles; defines the sixteen core edition order.
MANUSCRIPT_OPTIONAL_FILENAMES	Tuple listing `S01`, `S02`, `S03`, and `98_*` files included in stats and documentation beyond core inventory.
discover_manuscript_files	Infrastructure function sorting numeric stems, then `S*`, then `98_*`, then `99_*` references.

The sixteen core folios are the "edition pages"; `S01_*`–`S03_*` and `98_*` files document the system without changing the numbering of those pages. When adding optional slices, extend `MANUSCRIPT_OPTIONAL_FILENAMES` and update tests that assert file counts.

20.1 Narrative: how the terms fit together

A **folio** in this project is both a file and a pagination boundary. When `PDFRenderer` concatenates markdown, it inserts `\newpage` between discovered files so each slice begins a fresh page in the combined document (subject to LaTeX's float and widow rules). That is looser than historical newspaper "folio" usage, which sometimes counted leaves in a signature, but the metaphor holds: one file, one primary page break intent.

A **slice** is the same unit under a different metaphor—cutting the edition into horizontal layers you can reorder only by changing discovery order or renaming stems. Slices are not nested; there are no sub-slices inside `03_world.md` unless you manually simulate them with internal headings. Academic papers sometimes use `\section` commands generated by Pandoc; this project mostly uses markdown headings and raw LaTeX for column wrappers.

The **preamble** is not a slice. It is metadata injected around the document skeleton: paper size, packages, hyperref colors, and `multicol` parameters. Changing the preamble affects every page, whereas editing `11_local.md` affects only that folio's body (plus global dependencies like bibliography if you introduce conflicting packages). Keep preamble changes small and tested; they are the highest blast-radius edits.

Raw LaTeX blocks are escape hatches. Pandoc's markdown model does not know about `multicols` or `figure` environments, so this exemplar uses fenced `{=latex}` regions. The cost is that markdown linters may not analyze inside those fences; LaTeX becomes the authority. When a build fails, read the TeX log, not just the markdown validator output.

Thin orchestrators appear under `scripts/`. They set `sys.path`, call `get_logger`, invoke one or two functions from `src/newspaper/`, and print output paths for manifest collectors. If you feel tempted to add business logic in a script, move it into `src/` and cover it with tests—this pattern is constitutional for the monorepo.

Fixture copy functions exist so tests can generate paragraphs without scraping the web. Manuscript prose in this edition is mostly hand-written now, but `fixture_paragraph` remains useful when generating stress text programmatically or when teaching RNG determinism with NumPy seeds.

Figures, **captions**, and **labels** form a triad. The float environment owns the visual anchor; the caption explains it; the label lets prose reference "Figure 3" without hard-coding numbers. HTML export depends on captions being parse-friendly—avoid nested braces inside `\caption{}` text because the conversion filter uses a simple regular expression.

The **combined PDF** is the integration artifact. Individual slice PDFs help debugging, but reviewers usually want the stitched book. Paths vary slightly by renderer configuration; consult `output/traditional_newspaper/pdf/` after a successful run and look for `traditional_newspaper_combined.pdf` or similarly named outputs per your local template version.

XeLaTeX handles Unicode font paths better than pdfLaTeX for many modern workflows. If you switch engines, revisit `preamble.md` for font packages and compile twice (or more) whenever bibliography or references change.

layout_schematic.png encodes numbers from `NewspaperLayout`. When you change `column_sep_in` or margins, regenerate the PNG and update supplement copy if narrative descriptions drift from reality. Tests compare bytes for determinism; visual review confirms semantics.

PAGE_SLICES is the authoritative order of core news folios. Git blame it when someone asks why classifieds are last—because the exemplar models a full edition arc, not because classifieds are inherently final in all publications.

MANUSCRIPT_OPTIONAL_FILENAMES extends stats and documentation without participating in the sixteen-slice contract. Forgetting to append a new `S04` file here means `report_manuscript_stats.py` might skip it silently aside from discovery logs—keep the tuple synchronized.

discover_manuscript_files is infrastructure shared across projects. It excludes `preamble.md`, `AGENTS.md`, `README.md`, and config files so documentation can live beside manuscripts without polluting PDFs. If you add a new non-manuscript markdown helper, add its basename to the exclude set upstream rather than hacking per-project filters.

20.2 Using this glossary in reviews

When reviewing a pull request that touches `traditional_newspaper`, scan diffs for: stem renames (inventory tests), preamble edits (global risk), new floats (caption brace safety), and optional filename list updates (stats drift). Ask whether prose changes require regenerating figures or refreshing `manuscript_stats.json`. If the answer is yes, include those artifacts in the same commit series to keep `main` reproducible.

When onboarding a collaborator, assign reading order: root `README`, this glossary, `S01` supplement, then skim folios `01`–`16` to feel desk tone. Finally open `src/newspaper/AGENTS.md` for API tables. That path moves from product to implementation without drowning newcomers in LaTeX logs on day one.

20.3 Future-facing notes

Future maintainers may add `S04` slices for internationalization playbooks, `S05` for CI diagrams, or `99_references.md` if bibliography pages must be explicit. Each addition should update `MANUSCRIPT_OPTIONAL_FILENAMES`, bump any `len(files) >= N` tests, and document filenames in `manuscript/AGENTS.md`. Skipping those steps creates silent drift where stats JSON omits new prose even though PDFs include it—confusing anyone automating length gates.

If the edition grows beyond tabloid stress limits, consider splitting into morning and evening "editions" as separate projects under `projects/` rather than inflating one `PAGE_SLICES` tuple forever. The template's multi-project mode already lists several exemplars; cloning the newspaper pattern is cheaper than special-casing a thirty-slice megafile in discovery code.

Accessibility improvements might add alt text parameters to figure-generation functions, propagating into matplotlib `savefig` metadata and LaTeX `alt` macros where supported. That work belongs in `src/` with tests asserting non-empty strings for each public figure.

Security improvements might sandbox LaTeX compilation or validate included graphics are actually PNG/JPEG magic bytes. Those cross into infrastructure modules shared by all projects; propose changes upstream rather than forking locally without coordination.

Performance improvements might parallelize slice PDFs where references are local—careful: global bibliography and cross-slice references break naive parallelism. The combined build remains the source of truth until a graph-based planner understands reference edges.

Finally, keep this glossary honest: when terms change meaning after refactors, edit rows before the next release tag. Outdated glossaries erode trust faster than outdated jokes on the sports page.

References

Docxology. Research project template: Newspaper layout exemplar. https://github.com/docxology/template, 2026. Demonstration folio; not a peer-reviewed publication.